

NanoGT Match Report: Crigler-Najjar syndrome type I

Disease: Crigler-Najjar syndrome type I (ORPHA:1060)

Primary gene: UGT1A1

Gene CDS: 1596 bp

Inheritance: Autosomal recessive

Target tissues scored: liver

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility; AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone; Vector does not naturally cover all annotated disease tissues: liver.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase

Gene-addition compatibility: compatible

Preferred modality class: liver gene addition

Evidence level/status: direct / source_linked_needs_review

Evidence summary: UGT1A1 deficiency supports hepatocyte-directed gene addition

Evidence source: OMIM UGT1A1 gene entry

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint readiness: liver/metabolic targets may have biochemical biomarkers, but biomarker correction must be linked to clinical benefit.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Hemgenix	AAV5	7.2/10	Medium	approved
2	Roctavian	AAV5	7.2/10	Medium	approved
3	BMN 307	AAV5	7.2/10	Medium	phase2
4	Skysona	LV	7.1/10	Medium	approved
5	SPK-8011	AAVrh10	6.9/10	Medium	phase3

Match #1: Hemgenix

Precedent disease: Hemophilia B

Vector: AAV5

Tissue target: liver

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.24	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1596bp / cargo 4700bp (34% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: UGT1A1 deficiency supports hepatocyte-directed gene addition
- Mechanism source: OMIM UGT1A1 gene entry (<https://omim.org/entry/191740>)

- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — UGT1A1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Roctavian

Precedent disease: Hemophilia A

Vector: AAV5

Tissue target: liver

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.24	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1596bp / cargo 4700bp (34% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: UGT1A1 deficiency supports hepatocyte-directed gene addition
- Mechanism source: OMIM UGT1A1 gene entry (<https://omim.org/entry/191740>)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — UGT1A1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.19	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1596bp / cargo 4700bp (34% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: UGT1A1 deficiency supports hepatocyte-directed gene addition
- Mechanism source: OMIM UGT1A1 gene entry (<https://omim.org/entry/191740>)

- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — UGT1A1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.14	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1596bp / cargo 8000bp (20% utilized)
- No tissue overlap (disease: ['liver'], vector: ['hematopoietic'])
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: UGT1A1 deficiency supports hepatocyte-directed gene addition
- Mechanism source: OMIM UGT1A1 gene entry (<https://omim.org/entry/191740>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — UGT1A1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: liver
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

Match #5: SPK-8011

Precedent disease: Hemophilia A

Vector: AAVrh10

Tissue target: liver

Composite score: 6.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.90	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1596bp / cargo 4700bp (34% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Absent or severe deficiency of bilirubin UDP-glucuronosyltransferase
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: UGT1A1 deficiency supports hepatocyte-directed gene addition
- Mechanism source: OMIM UGT1A1 gene entry (<https://omim.org/entry/191740>)

- Approval status: phase3
- Vector immunogenicity (AAVrh10): moderate (~10%) — significant proportion may require pre-screening or exclusion
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — UGT1A1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone